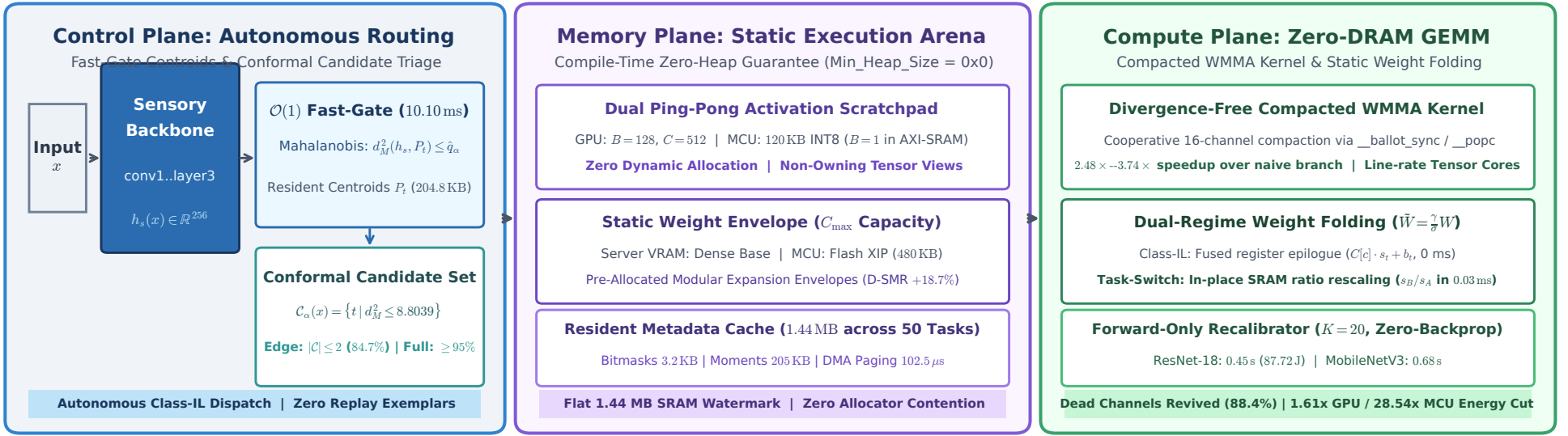

Control Plane: Autonomous Routing
Fast-Gate Centroids & Conformal Candidate Triage
Input
x
Sensory Backbone
conv1..layer3
$h_s(x) \in \mathbb{R}^{256}$
$\mathcal{O}(1)$ Fast-Gate (10.10 ms)
Mahalanobis: $d_M^2(h_s, P_t) \le \hat{q}_\alpha$
Resident Centroids P_t (204.8 KB)
Conformal Candidate Set
$\mathcal{C}_\alpha(x) = \{t \mid d_M^2 \le 8.8039\}$
Edge: $|\mathcal{C}| \le 2$ (84.7%) | Full: $\ge 95\%$
Autonomous Class-IL Dispatch | Zero Replay Exemplars
Memory Plane: Static Execution Arena
Compile-Time Zero-Heap Guarantee (Min_Heap_Size = 0x0)
Dual Ping-Pong Activation Scratchpad
GPU: $B = 128$, $C = 512$ | MCU: 120 KB INT8 ($B = 1$ in AXI-SRAM)
Zero Dynamic Allocation | Non-Owning Tensor Views
Static Weight Envelope ($C_{\max}$ Capacity)
Server VRAM: Dense Base | MCU: Flash XIP (480 KB)
Pre-Allocated Modular Expansion Envelopes (D-SMR +18.7%)
Resident Metadata Cache (1.44 MB across 50 Tasks)
Bitmasks 3.2 KB | Moments 205 KB | DMA Paging 102.5 µs
Flat 1.44 MB SRAM Watermark | Zero Allocator Contention
Compute Plane: Zero-DRAM GEMM
Compacted WMMA Kernel & Static Weight Folding
Divergence-Free Compacted WMMA Kernel
Cooperative 16-channel compaction via __ballot_sync / __popc
2.48× –3.74× speedup over naive branch | Line-rate Tensor Cores
Dual-Regime Weight Folding ($\tilde{W} = \frac{\gamma}{\sigma} W$)
Class-IL: Fused register epilogue ($C[c] \cdot s_t + b_t$, 0 ms)
Task-Switch: In-place SRAM ratio rescaling (s_B / s_A in 0.03 ms)
Forward-Only Recalibrator ($K = 20$, Zero-Backprop)
ResNet-18: 0.45 s (87.72 J) | MobileNetV3: 0.68 s
Dead Channels Revived (88.4%) | 1.61x GPU / 28.54x MCU Energy Cut